Monohybrid Inheritance

Exam code: 0610 & 0970

Download PDF

Written by: Phil
Reviewed by: Lára Marie McIvor
Updated on 23 January 2025

SaveMyExams

Did this video help you? Yes No

Inheritance: definitions

Help me understand

- **Inheritance** is the transmission of genetic information from one generation to the next generation
- A **gene** is a short length of DNA found on a chromosome that codes for a particular characteristic
- **Alleles** are versions, or forms, of a gene
  - Chromosomes exist in matching pairs, so individuals have two copies of each gene and therefore two alleles of each gene
    - One of the alleles is inherited from the mother and the other from the father
  - For example, an individual has two copies of the gene for eye colour; these alleles could be identical, or they could be different
- The observable characteristics of an organism is called the **phenotype**
- The combination of alleles that control each characteristic is called the **genotype**
- Alleles can be **dominant** or **recessive**
  - A dominant allele **only needs to be inherited from one parent** in order for the characteristic to show up in the phenotype
  - A recessive allele needs to be **inherited from both parents** in order for the characteristic to show up in the phenotype.

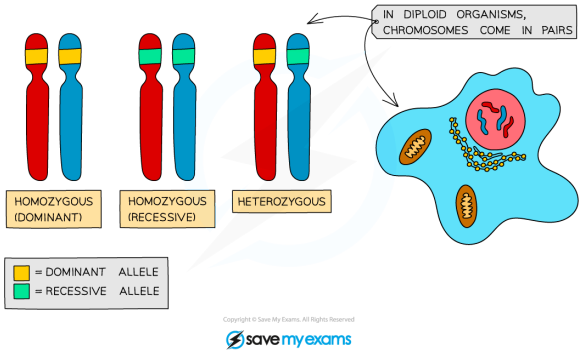
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

individual as being **homozygous** (homo = same); an individual could be:
    - **homozygous dominant**: having two copies of the dominant allele, or
    - **homozygous recessive**: having two copies of the recessive allele

- If the two alleles of a gene are different we describe the individual as being **heterozygous** (hetero = different)
- When completing genetic diagrams alleles are notated as single letters
    - The dominant allele is given a **capital letter** and the recessive allele is given the **lower case** version of the same letter



Alleles of a gene can carry the same instructions or different instructions. You can only inherit two alleles for each gene, and they can be the same or different

- We **cannot always determine the genotype** of an individual just by looking at the phenotype
    - A phenotype associated with a dominant allele will be seen in both a dominant homozygous and a heterozygous genotype

- If two individuals who are both **identically homozygous** for a particular characteristic are bred together, they will **produce offspring with exactly the same genotype and phenotype as the parents**
    - We describe them as **pure breeding** as they will always produce offspring with the same characteristics

- A **heterozygous** individual can pass on **different alleles** for the same characteristic each time it breeds with any other individual and can therefore **produce offspring with a different genotype and phenotype than the parents**
    - heterozygous individuals are not pure breeding

Genetic diagrams

What is monohybrid inheritance?

a **Punnett square**

- A Punnett square diagram shows the **possible combinations of alleles** that could be produced in the offspring
- From this the **ratio** of these combinations can be worked out

Worked Example

The height of pea plants is controlled by a single gene that has two alleles: tall and short

- The tall allele is dominant and is shown as **T**
- The small allele is recessive and is shown as **t**

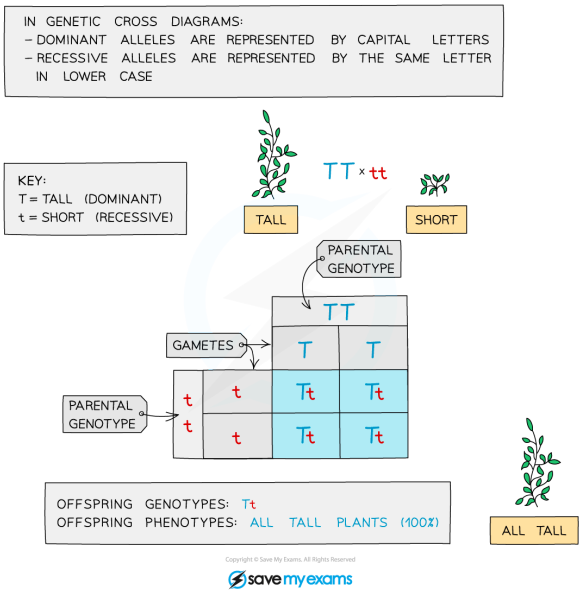
Show the possible allele combinations of the offspring produced when a homozygous short plant is bred with a homozygous tall plant. Determine the probability that any offspring will be tall

Step 1: construct a Punnett square

- The parents are homozygous, so:
  - tall = **TT**
  - short = **tt**
- The Punnet square should indicate:
  - parent gametes
  - offspring genotypes
  - an indication of which offspring are tall

Step 2: determine the probability of tall offspring

- All offspring are **Tt**
- The probability that they are tall = **100 %**



Copyright © Save My Exams. All Rights Reserved
save my exams

will be short.

Step 1: construct a Punnett square

- The offspring from the first cross are all **Tt**
- The Punnet square should indicate:
  - parent gametes
  - offspring genotypes
  - an indication of which offspring are short

Step 2: determine the probability of short offspring

- Offspring genotypes = **TT**, **Tt**, **Tt**, **tt**
- Offspring phenotypes = 3 x tall, 1 x short
- The probability that they are tall = **25 %**

Worked Example

The height of pea plants is controlled by a single gene that has two alleles: tall and short

- The tall allele is dominant and is shown as **T**
- The small allele is recessive and is shown as **t**

Show the results of crossing a heterozygous plant with a short plant. Determine the probability of the offspring being short.

Step 1: construct a Punnett square

- The heterozygous parent has the genotype **Tt**
- The short parent has the genotype **tt**
- The Punnet square should indicate:
  - parent gametes
  - offspring genotypes
  - an indication of which offspring are short

- Offspring phenotypes = 2 x tall, 2 x short
- The probability that they are tall = **50 %**

Constructing Punnett squares

- You may be asked to determine the ratio of different allele combinations in the offspring, calculate a percentage chances of offspring showing a specific characteristic or just determine the phenotypes of the offspring; this can be done using a Punnett square as follows:
  1. Determine the letter that will be used to notate each allele
     - Some exam Qs will provide this information
     - If not then you should select a relevant letter, e.g. E and e for **e**ye colour
  2. Determine the parental genotypes
  3. Determine the gametes produced by each parent and add them to the Punnett square headings
  4. Fill in the middle four squares of the Punnett square to work out the possible genetic combinations in the offspring
  5. Indicate clearly on your Punnett square which individual will show each phenotype, e.g. with labels or colour coding

Examiner Tips and Tricks

If you are asked to use your own letters to represent the alleles in a Punnett square, try to choose a letter that is obviously different as a capital than the lower case so the examiner is not left in any doubt as to which is dominant and which is recessive.

Pedigree diagrams

- Family pedigree diagrams are usually used to trace the **pattern of inheritance** of a specific characteristic (usually a disease) **through generations of a family**
- This can be used to work out the probability that someone in the family will inherit the genetic disorder

A family pedigree chart

- Males are indicated by the **square shape** and females are represented by **circles**
- Affected individuals are **red** and unaffected are **blue**
- Horizontal lines between males and females show that they have produced children (which are shown underneath each couple)
- The family pedigree above shows:
  - both males and females are affected
  - every generation has affected individuals
  - That there is one family group that has no affected parents or children
  - the other two families have one affected parent and affected children as well

Identifying an unknown genotype: extended

- Breeders can use a **test cross** to find out the genotype of an organism showing the dominant phenotype
- This involves crossing the unknown individual with an individual showing the recessive phenotype - if the individual is showing the recessive phenotype, then its genotype must be homozygous recessive
- By looking at the **ratio of phenotypes in the offspring**, we can tell whether the unknown individual is homozygous dominant or heterozygous

The height of pea plants is controlled by a single gene that has two alleles: tall and short

- The tall allele is dominant and is shown as **T**
- The small allele is recessive and is shown as **t**

A plant breeder has a tall plant of unknown genotype. Show how they can find out whether it is homozygous dominant or heterozygous.

Step 1: construct Punnett squares

- The short plant is showing the recessive phenotype and so must be homozygous recessive **tt**
- The tall parent could be **either TT or Tt**, so we will need to produce two Punnett squares to show the different outcomes
- The Punnet squares should indicate:
  - parent gametes
  - offspring genotypes
  - which offspring are tall or short

Step 2: determine how the outcomes of the two crosses would differ

- Homozygous tall parent offspring = Tt = all tall
- Heterozygous tall parent offspring genotypes = Tt or tt = 50 % tall and 50 % short
- **The presence of any short offspring indicate that the unknown parent has a heterozygous genotype**

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.